

Waste Hotspot Summary for Nora High School

A Sustainability Assessment to Reduce Energy Costs and Waste

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Executive Summary

This preliminary audit shows sustainability opportunities at Nora High School, a private secondary school in Garki, Abuja. The school seeks to reduce operational costs and align with government sustainability mandates through smarter energy use, water management, and waste reduction. Focusing on four key zones, which are Dormitories, Administrative Block, Classrooms, and Cafeteria, the audit identifies high-impact areas (waste hotspots), highlights inefficiencies, and proposes practical, scalable recommendations. The goal is to support Nora High in adopting greener practices while building a culture of environmental awareness among students and staff.

Audit Scope & Purpose

This purpose for this audit was to help Nora High:

- Understand its current energy, water, and waste footprint.
- Identify inefficiencies and potential cost savings.
- Align with government directives encouraging sustainable practices in educational institutions.
- Educate its community on environmental responsibility and foster a culture of sustainability.

Methodology

The audit was conducted through scenario-based analysis guided by regional energy patterns, typical school infrastructure, and sustainability frameworks. Observations and recommendations were modeled on real-life use cases and facility types (e.g., boarding dorms, cafeterias, classrooms) commonly found in urban secondary schools in Nigeria. No physical audits were performed; this report is designed as a learning tool and portfolio sample.

Audit Areas + Key Findings

AREA A: Dormitory

The dormitory at Nora High accommodates day-to-day living needs of boarding students and is a key area of energy and water consumption. It's a simple, functional setup, with some opportunity for sustainable upgrades.

Observations:

i. Ventilation & Cooling:

The dormitory uses ceiling fans and open windows for ventilation. There are no air conditioners, but a few students privately own table fans due to recent heat waves.

ii. Lighting:

Most dorm rooms rely on tube lights or standard bulbs. A typical room may have two bulbs depending on the number of occupants. No motion sensors or automatic lighting controls are in place.

iii. Electric Appliance Use:

Use of personal electrical appliances (e.g. water heaters or pressing irons) is generally restricted. However, during generator hours or when public electricity is available, some students use personal fans or occasionally plug in irons under supervision.

iv. Water System:

Water is supplied through boreholes and stored in overhead tanks. Bathrooms are fitted with showers and basic storage containers. Water pressure can vary, especially during dry seasons.

v. Waste Generation:

The dormitory generates significant plastic waste from bottled water, sachet drinks, and food wrappers. Paper waste is also common. While energy waste is low due to controlled generator hours (typically 7–10 p.m.), students are not routinely trained in energy conservation. Waste separation or recycling is not currently practiced.

AREA B: Admin Block

The Administrative Block of Nora High accommodates core school leadership and operational staff. This includes offices for the Proprietor/Proprietress, Principal, Vice Principals (Academic and Administration), Record Room, Bursar's Office, and the Staff Room for teachers.

Lighting and Cooling:

Each main office (especially those of the Principal, VPs, and Proprietor) is fitted with standard LED lighting and energy-intensive cooling units (ACs). The staff room and record rooms utilize tube lights and ceiling fans. While these installations ensure comfort, they significantly contribute to the school's electricity demand.

Current Sustainable Features:

Administrative offices make partial use of solar energy, helping to reduce dependence on the grid.

The school has designated waste collection zones within the admin block premises, including basic waste segregation practices (paper, plastic, biodegradables).

Key Observations:

- Offices with ACs typically lack automated timers or energy-efficient settings.
- Lighting remains active during daylight hours despite natural lighting potential.
- Electronic records and data are not consistently managed in a paperless format, increasing paper waste.

AREA C: Classrooms**i. Energy Use**

Classrooms at Nora High rely on the school's central energy supply, which may include grid power or generator backup depending on availability. In well-lit (Band A) areas of Abuja, electricity can be stable enough to run fans and lighting throughout the day. However, in less serviced areas, schools may restrict usage or prioritize admin blocks and essential services. Classrooms typically use:

- 1–2 ceiling fans per room
- 1–2 fluorescent bulbs per class
- No heavy-duty or high-consumption equipment is used in typical classrooms.

ii. Water Use

There is no direct water line to classrooms. Shared toilet and hand-washing stations exist outside classrooms, usually split by junior and senior levels.

iii. Equipment & Infrastructure

Most classrooms use basic marker boards, desks, and ceiling fans. Some high-end schools may adopt projectors or smart boards, but this is uncommon at the secondary school level in this context.

iv. Waste Management

Each class usually has one waste bin. However, waste segregation is not practiced; paper, plastic, and organic waste are often disposed of together. School cleaners typically do any sorting.

AREA D: Cafeteria

i. Energy Use

For safety and cost efficiency, Nora High's cafeteria uses gas for most of its cooking operations. The appliances observed or expected in the kitchen include:

- Gas burners (primary cooking method)
- Microwave ovens for reheating
- Electric ovens for baking
- Electric cookers (occasional use)

Cooling is primarily done with standing or table fans, as kitchens typically become hot. It is unlikely to have AC units in the cooking area, but staff may bring personal electric fans for comfort.

ii. Lighting

Basic ceiling bulbs or tube lights are used to provide adequate visibility during early morning and evening prep hours

iii. Water Use

Water usage in the kitchen is significant for cooking, cleaning, and washing produce and utensils. A dedicated water storage tank serves the kitchen. There's a strong case for introducing greywater recycling to reuse lightly used water (e.g. from rinsing) for non-potable purposes.

iv. Waste Generation & Management

The cafeteria is a major source of waste, including:

- Food scraps (plate leftovers, peels, expired items)
- Single-use packaging (nylon wraps, sachets, foil, plastic containers)
- Non-organic waste (damaged utensils, plastics)

Presently, no structured composting or food waste diversion is practiced. Waste is disposed of unsorted, representing a missed opportunity for waste reduction and environmental responsibility.

Waste Hotspot Summary

After observing the Dormitory, Cafeteria, Admin Block, and Classrooms, the following hotspots were identified:

Cafeteria (Primary Hotspot)

- High volume of food waste, plastic packaging, and energy-intensive cooking.
- No waste segregation or composting practice in place.

Dormitory

- Inconsistent energy use patterns (late-night lights, unmanaged fans).
- Plastic bottles and personal packaging waste common.

Admin Block

- High energy load from ACs, fans, and office electronics.
- Paper waste due to manual record-keeping and poor digital adoption.

Classrooms

- Unnecessary lighting left on.
- No bins per class, leading to improper disposal of packaging and stationery waste.

Waste Hotspot Matrix Table

Audit Area	Key Waste Types	Severity Level	Opportunities for Action	Priority
Cafeteria	Food waste, Plastic packaging, Disposable containers	● High	Composting, food-waste diversion, structured waste sorting	● High
Dormitory	Plastic bottles, Personal wrappers, Energy waste (lights/fans)	● Medium	Awareness campaigns, waste bin labeling, motion-sensor lights	● Medium
Admin Block	Paper waste (manual records), Electricity from ACs	● Medium	Digital records, AC policies, daylight optimization	● Medium
Classrooms	Mixed waste (paper/plastic), Lighting left on	● Low	Teacher-led awareness, individual bin systems	● Low

The severity levels are based on the volume, frequency and potential environmental impact

Recommendations

After careful evaluation and assessments, the following were proposed:

For Admin Block

- Install motion sensor lighting in low-traffic office zones and optimize natural daylight.
- Introduce strict AC usage policies (timer settings, energy rating compliance)
- Encourage digital record keeping to reduce paper usage.
- Formalize and upgrade the waste segregation system with labeled bins.

For Dormitory Block

- Install reminder stickers near switches to encourage lights-off behavior.
- Educate hostel managers and student leaders on energy-saving practices.
- Provide labeled waste bins to separate plastic, paper, and food waste.
- Install solar-powered motion-sensor lighting in shared areas (corridors, bathrooms).
- Introduce “sustainability stewards” within hostels to reinforce good habits.
- Explore options for a basic composting system or recycling partnership for collected waste.

For Classrooms

- Encourage teachers and prefects to assign “Energy Monitors” in each class responsible for turning off lights/fans when not in use.
- Place visual cues or posters near switches and bins to promote energy and waste awareness.
- Transition to energy-efficient LED bulbs across all classrooms.
- Introduce color-coded bins in each block to promote waste segregation from the source.

For Cafeteria

- Train kitchen staff to separate food waste and packaging waste at the point of disposal.
- Collaborate with local farmers or composting vendors to repurpose organic waste.
- Install basic greywater filtration to reuse water from rinsing and non-greasy cleaning.
- Pilot reusable packaging systems (metal trays, washable cups) for in-school food service.

Conclusion

Though Nora High is a fictional school, its challenges reflect real sustainability gaps in education facilities across Nigeria. This audit shows how even small institutions can make meaningful progress by applying basic principles of energy efficiency, water conservation, and waste management. With growing pressure to build climate-resilient systems, the practices outlined here can guide school leadership in building a greener, more cost-effective future, one that also nurtures the next generation of sustainability-conscious citizens.